EC3214 — International Finance

Summer 2022 Examination · Paper 1 · Full Marking Scheme

Module	EC3214 — International Finance
Internal Examiner	Ms. Aisling Finucane
Institution	University College Cork
Session	Summer 2022, Paper 1
Duration	1.5 hours
Structure	Section A — Long Questions: attempt **2 of 3** (100% of the paper). All questions carry equal weight.
Materials	Non-programmable calculator permitted

Coverage note. This scheme answers **all three** Section A questions in full, since you'll choose your best two under exam conditions.

Consistency with other sittings. A2 on this paper is worded essentially " "identically to the Summer 2023 EC3214 sitting, with the same numbers throughout — the underlying content below is word-for-word consistent with that scheme. A1(a) is also consistent with the Summer 2023/2024 sittings. A1(b) here asks about the same floating-rate mechanics as those sittings, but with an extra item — economic growth — alongside the familiar national/real income shock; see the note at that question on how the two relate.

Source basis. A3's discussion of "Dutch Disease" and its application to the UK draws on the standard economics literature (The Economist, 1977; the broader Dutch disease and "finance curse" literature applied specifically to the UK's financial-services-heavy economy) — flagged at that question since this wasn't set out in the uploaded course material.

Section A · Question 1

The flexible-price monetary model: assumptions and four comparative statics — 100 marks

A1(a) [40 marks]

Explain the assumptions underlying the flexible price monetary approach model.

A1(b) [60 marks]

Explain using the flexible price monetary approach model how each of the following will affect the home country exchange rate: (i) An expansion in the home country money supply. (ii) An increase in economic growth in the home country. (iii) A foreign price increase under floating exchange rates. (iv) A rise in real income in the domestic economy.

FULL MODEL ANSWER

- **Purchasing power parity holds continuously:** absolute PPP ($S = P/P^*$) is assumed to hold at every point in time, not just in the long run — the exchange rate is always exactly consistent with relative price levels.
- **Prices are fully flexible:** both domestic and foreign price levels adjust instantaneously to clear their respective money markets, with no stickiness of any kind.
- **Output is fixed at its full-employment level:** real income y is determined independently of monetary factors (by the economy's real productive capacity), so monetary shocks cannot move output — **money is neutral**.
- **Stable money demand function:** real money demand is a stable, well-behaved function of income (and, in extended versions, the interest rate), so that $M_o^s = kPy$ (or its interest-rate-augmented version) reliably describes money market equilibrium.
- **Perfect capital mobility / uncovered interest parity:** financial markets are assumed to be perfectly integrated internationally, so that interest rate differentials reflect expected exchange rate movements.

Mark Allocation — Part (a) (40 marks)

Tier	Marks	What separates this tier
31–40	31–40	At least four of the five core assumptions correctly stated and explained (continuous PPP, full price flexibility, fixed/full-employment output, stable money demand, perfect capital mobility/UIP).
21–30	21–30	At least three assumptions correctly stated.
11–20	11–20	Only one or two assumptions correctly stated, or stated without explanation.

0-10	**0-10**	No credible assumptions given.

EXAM TECHNIQUE

Parts (ii) and (iv) both concern a rise in domestic real output/income, and operate through **exactly the same mechanism** in this model — the model has no separate channel through which "economic growth" (a rate of change) differs from "a rise in real income" (a level change) in its qualitative prediction for the exchange rate. Rather than searching for a different mechanism for (ii), the strongest answer explicitly notes that both items test the same $y\uparrow$ channel and gives the identical, correctly-reasoned answer to both — this is worth more marks than inventing an artificial distinction that doesn't exist in the model.

Setting up the model: money market equilibrium gives $P = M_o^s/(ky)$; PPP gives $S = P/P^*$ (domestic currency units per unit of foreign currency — a **rise** in S is a domestic currency **depreciation**; a **fall** is an **appreciation**).

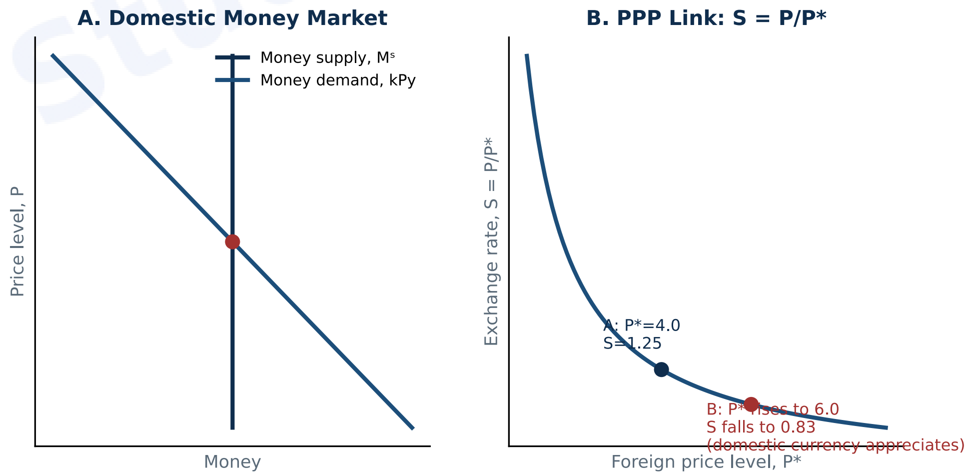

Fig. 1 — Panel A: the domestic money market determines P. Panel B: the PPP link maps P into S given P. Each shock below works by shifting one of these two relationships.*

Shock	Effect on P	Effect on S	Direction
(i) Home money supply expansion ($M_o^s\uparrow$)	P rises proportionally ($P=M_o^s/ky$)	S rises proportionally	Depreciation
(ii) Economic growth in the home country ($y\uparrow$)	P falls (money demand ky rises, M_o^s fixed)	S falls	Appreciation
(iii) Foreign price increase ($P^*\uparrow$), floating rates	No change (domestic side unaffected)	S falls (P^* in denominator)	Appreciation
(iv) Rise in domestic real income ($y\uparrow$) — same shock as (ii)	P falls, for the same reason as (ii)	S falls, for the same reason as (ii)	Appreciation

(i) Home money supply expansion: P rises proportionally with M_o^s (classic quantity-theory logic), and via PPP, S rises by the same proportion — the domestic currency depreciates in direct proportion to the money supply increase.

(ii) and (iv) — economic growth / a rise in real income (the same mechanism): a rise in y raises real money *demand* (ky). With the money *supply* unchanged, equilibrium requires the price level P to **fall** (domestic deflation) to bring real money balances (M_o^s/P) up to meet the higher demand.

Lower domestic prices, via PPP, mean S falls — the domestic currency **appreciates**. Whether this higher y is described as "economic growth" (the more dynamic framing) or simply "a rise in real income" (the direct comparative-static framing), the underlying shock to the money market and its qualitative effect on S are identical.

(iii) Foreign price increase, under floating rates: with nothing on the domestic side changing, a rise in P* directly appreciates the domestic currency through the PPP identity S = P/P* alone — P* rising with P fixed pushes S down.

Mark Allocation — Part (b) (60 marks)

Tier	Marks	What separates this tier
46–60	**46–60**	All four comparative statics correctly signed and explained via the $P=M_0^s/(ky)$ and $S=P/P^*$ mechanism **AND** parts (ii) and (iv) are explicitly recognised as the same underlying shock, rather than treated as requiring separate mechanisms **AND** the diagram correctly illustrates the mechanism.
31–45	**31–45**	At least three of the four correctly signed and explained, but (ii)/(iv) are treated as requiring different reasoning without justification.
16–30	**16–30**	Only two of the four correctly explained.
0–15	**0–15**	Fewer than two correctly explained, or no credible mechanism given for any of them.

Question A1 Total: 100 Marks

Section A · Question 2

PPP and the real exchange rate, and two cross-currency triangulations — 100 marks

A2(a) [40 marks]

Explain the purchasing power parity theory of exchange rate determination. What does this imply for the Real Exchange Rate?

A2(b) [30 marks]

Given the following interbank FX rates, what is the GBP/CHF rate? EUR/GBP 0.6642/45, EUR/CHF 1.5745/48.

A2(c) [30 marks]

Given the following interbank FX rates, what is the GBP/SGD rate? GBP/USD 1.6523/28, USD/SGD 1.7585/1.7600.

FULL MODEL ANSWER

Purchasing power parity theory: PPP holds that exchange rates adjust to equalise the purchasing power of different currencies — in its absolute form, $P = SP^*$, the price of an identical basket of goods should be the same everywhere once converted to a common currency via the exchange rate S. The theory rests on the **law of one price** generalised across an entire basket of goods: if a basket were cheaper in one country once converted to a common currency, arbitrageurs would buy there and sell where it's expensive, and this arbitrage activity — bidding up demand (and prices) where goods are cheap and increasing supply (pushing down prices) where they're expensive — would continue until the price gap closed.

The relationship: the real exchange rate is defined as $Q = SP^*/P$ — the nominal exchange rate S adjusted for relative price levels. Absolute PPP ($P = SP^*$) holds exactly if and only if $\mathbf{Q = 1}$: by direct substitution, $Q = SP^*/P = SP^*/(SP^*) = 1$. So PPP and the real exchange rate are two ways of describing the same underlying claim — PPP asserts that the real exchange rate is constant and equal to 1, meaning a given basket of goods costs the same everywhere once converted to a common currency, leaving no genuine difference in purchasing power once the nominal exchange rate is accounted for.

Why deviations of Q from 1 matter economically: when $Q \neq 1$, the real exchange rate is capturing genuine differences in relative competitiveness — a country whose real exchange rate has appreciated (Q risen, in a convention where higher Q means the country's goods are relatively more expensive once converted to a common currency) has become less price-competitive internationally, regardless of what its nominal exchange rate has done, since nominal exchange-rate movements can be offset or reinforced by relative inflation differences.

Mark Allocation — Part (a) (40 marks)

Tier	Marks	What separates this tier
31–40	**31–40**	Correctly states absolute PPP ($P=SP^*$) with the law-of-one-price foundation and arbitrage mechanism explained **AND** correctly derives Q=1 from PPP **AND** explains the economic meaning of the real exchange rate.
21–30	**21–30**	Correct PPP statement and Q=1 derivation, but the arbitrage mechanism underlying PPP is not explained.
11–20	**11–20**	Vague statement of PPP without the formula or the real exchange rate implication.
0–10	**0–10**	No credible answer given.

Setting up the triangulation: EUR is the common **base** currency in both quoted pairs (EUR/GBP and EUR/CHF), so the cross rate is built by dividing the two quotes so the EUR term cancels — using bid-with-offer, offer-with-bid matching to protect the dealer's margin on the synthetic transaction.

Quote	Calculation	Result
GBP/CHF bid	1.5745 ÷ 0.6645	2.3695

GBP/CHF offer	**1.5748 ÷ 0.6642**	**2.3710**

Result: the no-arbitrage GBP/CHF quote is **2.3695 / 2.3710**.

Mark Allocation — Part (b) (30 marks)

Tier	Marks	What separates this tier
23–30	**23–30**	Correctly identifies EUR as the common base currency to cancel **AND** correctly applies bid-with-offer, offer-with-bid triangulation **AND** arrives at 2.3695 / 2.3710 (or a consistent own-figure result).
15–22	**15–22**	Correct triangulation method, but an arithmetic error in the final figures.
8–14	**8–14**	Incorrect method (e.g. bid-with-bid) but carried through consistently.
0–7	**0–7**	No coherent triangulation attempted.

Setting up the triangulation — a direct chain, not a common-base cross: here USD is the **quote** currency in GBP/USD but the **base** currency in USD/SGD — a direct chain (GBP → USD → SGD). A direct chain like this is built by **multiplying** the two rates (matching bid-with-bid, offer-with-offer), not dividing.

Quote	Calculation	Result
GBP/SGD bid	1.6523 × 1.7585	2.9056
GBP/SGD offer	**1.6528 × 1.7600**	**2.9089**

Result: the no-arbitrage GBP/SGD quote is **2.9056 / 2.9089**.

EXAM TECHNIQUE

This question and the previous one are deliberately structured to require two different triangulation techniques — a common-base cross (divide, cross bid with offer) versus a direct chain (multiply, match bid with bid and offer with offer). Identifying which structure you're facing before choosing a method is the first and most important step in both.

Mark Allocation — Part (c) (30 marks)

Tier	Marks	What separates this tier
23–30	**23–30**	Correctly identifies this as a direct chain (USD is quote in one pair, base in the other) requiring multiplication rather than division **AND** correctly matches bid-with-bid, offer-with-offer **AND** arrives at 2.9056 / 2.9089 (or a consistent own-figure result).
15–22	**15–22**	Correct multiplicative method, but an arithmetic error in the final figures, or bid/offer sides mismatched.
8–14	**8–14**	Incorrectly applies Part (b)'s division method to this chain structure, but carries the (wrong) method through consistently.
0–7	**0–7**	No coherent triangulation attempted.

Question A2 Total: 100 Marks

Section A · Question 3

Dutch Disease, and its application to pre-Brexit Britain — 100 marks

A3(a) **[40 marks]**

What is your understanding of "Dutch Disease" in international finance?

A3(b) **[60 marks]**

"Pre-Brexit, Britain was obviously experiencing a version of so-called Dutch Disease." Based on your reading of the literature, discuss the above statement.

FULL MODEL ANSWER

EXAM TECHNIQUE

"Dutch Disease" wasn't set out in the uploaded course material. The explanation below follows the standard treatment, sourced to its origin (*The Economist*, 1977) and the subsequent development economics and international finance literature.

Origin and definition: the term was coined by *The Economist* in 1977 to describe the decline of Dutch manufacturing following the discovery of the large Groningen natural gas field in 1959. "Dutch Disease" describes the **apparent causal relationship between a boom in one sector of an economy (classically, natural resource exports) and a decline in other sectors** (classically, manufacturing) — an economy that appears to be getting richer from a windfall ends up structurally worse off, because the windfall itself damages the competitiveness of the rest of the tradable economy.

Dutch Disease: Two Channels from a Boom to Deindustrialisation

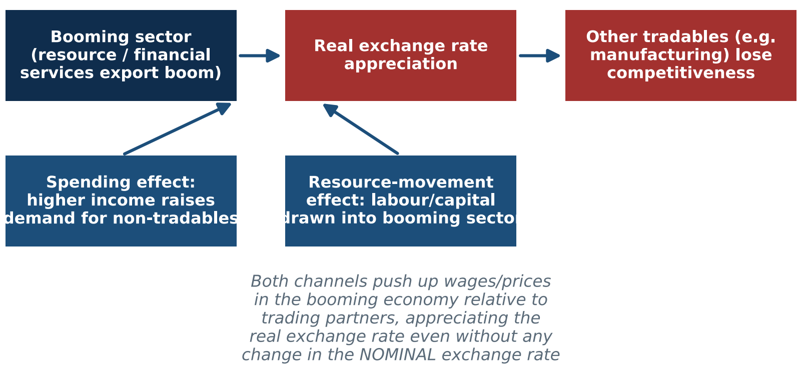

Fig. 1 — Dutch Disease operates through two channels: the spending effect (higher income raises demand for non-tradables, bidding up their prices) and the resource-movement effect (labour and capital are drawn into the booming sector) — both of which appreciate the real exchange rate and squeeze other tradables' competitiveness.

The two transmission channels

- **The spending effect:** the booming sector generates a surge in national income, which raises domestic demand — including demand for **non-tradable** goods and services (housing, local services) whose supply cannot be met by imports. Prices and wages in the non-tradable sector are bid up, which — because wages and costs are shared across the whole economy — raises costs for the tradable (manufacturing) sector too, eroding its international competitiveness even without any change in the nominal exchange rate.
- **The resource-movement effect:** the booming sector, now paying higher wages, draws labour and capital directly away from other tradable sectors (like manufacturing) and into itself, shrinking those other sectors' productive capacity directly.
- **The exchange-rate channel (where the boom generates foreign currency inflows):** for a resource-export boom specifically, foreign buyers need the domestic currency to purchase the resource, which appreciates the **nominal** exchange rate directly, compounding the real appreciation from the two channels above and making every other tradable good more expensive to foreign buyers and imports cheaper at home.

> **Why this matters beyond the immediate windfall:** the long-run concern is that manufacturing and other tradable sectors often generate more valuable, durable spillovers (learning-by-doing, innovation, export capability) than the booming sector does — so a temporary boom can permanently hollow out productive capacity that is costly or impossible to rebuild once the boom itself fades, which is why Dutch Disease is treated as a genuine long-run structural risk, not merely a short-run relative-price adjustment.

Mark Allocation — Part (a) (40 marks)

Tier	Marks	What separates this tier
31–40	**31–40**	Correct origin and definition given **AND** at least two of the three transmission channels (spending effect, resource-movement effect, direct exchange-rate effect) correctly explained **AND** the long-run structural concern (irreversible loss of tradable-sector capacity) is articulated.
21–30	**21–30**	Correct definition with at least one channel correctly explained.
11–20	**11–20**	Vague definition ("a resource boom hurts other industries") without a clear mechanism.
0–10	**0–10**	No credible answer given.

> **The core of the argument:** a substantial strand of the literature — sometimes termed **"financial Dutch disease"** or the **"finance curse"** — argues that the UK's outsized, globally dominant financial services sector (the City of London) played precisely the role a natural-resource sector plays in the classical model, with broadly the same consequences for the rest of the UK's tradable economy, particularly manufacturing.

How the mechanism maps onto pre-Brexit Britain

- **Large capital inflows propping up sterling:** the City's role as a global financial centre attracted enormous, persistent inflows of foreign capital — much of it, critics note, directed toward mergers and acquisitions and property purchases rather than productive new investment — which kept sterling's value elevated relative to what the competitiveness of UK manufacturing alone would have justified. By 2015, Deutsche Bank had reportedly called sterling the most overvalued major currency in the world.
- **The resource-movement effect, applied to human capital:** the finance sector's exceptionally high salaries drew a disproportionate share of graduates and skilled workers away from manufacturing, engineering, and other tradable sectors — a direct parallel to the classical resource-movement channel, but

operating through the labour market for skilled workers rather than raw factor inputs.

- **The spending effect, via London property and services:** financial-sector wealth concentrated heavily in London and the South East drove sharp increases in property prices and local service costs, raising the general cost base UK-wide manufacturers had to compete against, in a pattern commentators have directly likened to the classical Dutch Disease spending-effect channel.
- **The long-run consequence — persistent deindustrialisation:** critics of this "financialised growth model" argue that decades of this dynamic left UK manufacturing and exporting "a shadow of its former self" well before Brexit itself — meaning Brexit-era currency moves (sterling's post-referendum fall) could not simply restore manufacturing competitiveness overnight, since the underlying productive capacity, supply chains, and skills base had already been eroded over a much longer period.

> **A structural nuance worth engaging with — Krugman's own framing:** commentators have specifically invoked Dutch Disease reasoning around the Brexit referendum itself — most notably, Paul Krugman argued in its aftermath that a falling pound could actually be beneficial for UK manufacturing, precisely because it would partially unwind the currency overvaluation the finance-driven boom had generated. This framing treats the pre-Brexit overvaluation as the *direct symptom* of financial Dutch disease, with a weaker post-referendum pound representing a (partial, and contested) natural correction.

> **A balanced assessment:** the "financial Dutch disease" reading of pre-Brexit Britain is a genuine, actively-debated position in the literature, not a settled consensus — the mechanism (financial-sector capital inflows and wage effects appreciating the currency and squeezing manufacturing) is coherent and consistent with the classical model, and is supported by the persistent overvaluation evidence and the specific commentary cited above. It is best treated as a genuinely strong and well-evidenced *partial* explanation for the UK's long-run manufacturing decline and sterling overvaluation, sitting alongside — not necessarily displacing — other standard explanations for deindustrialisation (global trade competition, automation, policy choices around industrial strategy), rather than a single, complete account on its own.

Mark Allocation — Part (b) (60 marks)

Tier	Marks	What separates this tier
46–60	**46–60**	Correctly identifies the "financial Dutch disease"/finance curse argument and maps at least two of its specific channels onto the UK's financial-sector-driven pre-Brexit economy **AND** engages with specific supporting evidence (sterling overvaluation, the finance-sector brain-drain effect) **AND** gives a balanced assessment that treats this as a strong partial explanation rather than either dismissing it or treating it as the sole/complete cause.
31–45	**31–45**	Correctly identifies the financial Dutch disease argument with at least one channel mapped onto the UK case, but without specific supporting evidence or a balanced assessment.
16–30	**16–30**	Discusses Dutch Disease in general terms without specifically applying it to the UK's financial-sector-driven economy.
0–15	**0–15**	No credible discussion of the statement given.

Question A3 Total: 100 Marks